



Changchang Yin

Ph.D. Candidate

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Bio

- I am a research scientist in the Department of Biomedical Informatics at The Ohio State University (OSU). I received BS and MS in Computer Science and Technology from Xi'an Jiaotong University in 2016 and 2019, and PhD in Computer Science from Computer Science and Engineering (CSE), OSU, working with Dr. Ping Zhang. My research interests lie in data mining, machine learning, deep learning, and their application to trustworthy AI (e.g., causal inference, model interpretability, data privacy & security, and model uncertainty estimation), and computational medicine (e.g., predictive modeling, patient subtyping, dynamic treatment regime learning, and medical imaging). My research findings (including 8 first-author papers, 5 co-first-author papers, and 20+ co-author papers) have been published at top-tier venues such as KDD, ICDM, AAAI, IJCAI, ACL-IJCNL, CHI, Nature Computational Science, and JMIR, achieving more than 3,600 citations.

EDUCATION

- Ph.D. The Ohio State University, Columbus, OH, USA**
Major: Computer Science and Engineering 2019 - 2024
Advisor: Dr. Ping Zhang
- M.S. Xi'an Jiaotong University, Xi'an, Shaanxi, China**
Major: Computer Science and Technology 2016 - 2019
Advisor: Dr. Buyue Qian
- B.S. Xi'an Jiaotong University, Xi'an, Shaanxi, China**
Major: Computer Science and Technology 2012 - 2016

PROFESSIONAL EXPERIENCE

Department of Biomedical Informatics, The Ohio State University **Columbus, OH, USA**
Research Scientist 2025 - Present

Artificial Intelligence in Medicine (AIMed) Lab, The Ohio State University **Columbus, OH, USA**
Graduate Research Assistant 2019 - 2024

- Advisor: Dr. Ping Zhang
- Topic: Trustworthy Deep Learning on Electronic Health Records
 - Interpretability: developed interpretable risk prediction models with attention mechanisms and an interactive clinical prediction system to show which clinical records could increase or decrease the disease risks and what if the early intervention (e.g., medication) is assigned (*ICDM 2019* and *JMIR 2020*)
 - Privacy & Security: developed federated treatment effect estimation model on decentralized data, which addresses both global and local confounding bias issues with two weights (*Nature Communications 2024 submission*)
 - Causality: developed a deconfounding actor-critic model to learn optimal personalized treatment strategies from observational EHRs, which incorporated a patient resampling module and a confounding balance module to remove confounding bias (*KDD 2022*)
 - Uncertainty: developed an active sensing model to quantify the clinical risk prediction models' uncertainty caused by the unobserved variables and recommend new observations to reduce the uncertainty (*KDD 2024*)

The full list of my papers is available at Google Scholar (3,600+ citations, h-index: 16, and i10-index: 19). Note that * denotes co-first authors.

1. Siyi Wu, Weidan Cao, Shihan Fu, Bingsheng Yao, Ziqi Yang, **Changchang Yin**, Varun Mishra, Daniel Addison, Ping Zhang, Dakuo Wang. CardioAI: A multimodal AI-based system to support symptom monitoring and risk prediction of cancer treatment-induced cardiotoxicity. ACM CHI Conference on Human Factors in Computing Systems (**CHI**), 2025 (Acceptance rate: 1249/4976 = 25.1%)
2. **Changchang Yin**, Shihan Fu, Bingsheng Yao, Thai-Hoang Pham, Weidan Cao, Dakuo Wang, Jeffrey Caterino, Ping Zhang. SepsisCalc: Integrating clinical calculators into early sepsis prediction via dynamic temporal graph construction. ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**), 2025
3. Jin Peng, Jiayuan Chen, **Changchang Yin**, Ping Zhang, Jingzhen Yang. Comparison of machine learning models in predicting mental health sequelae following concussion in youth. American Medical Informatics Association Informatics Summit (**AMIA Summit**), 2025
4. Zishan Gu, Fenglin Liu, Jiayuan Chen, **Changchang Yin**, Ping Zhang. A proactive agent collaborative framework for zero-shot multimodal medical reasoning. Advanced Intelligent Systems, 2025 (Impact factor: 6.8)
5. Thai-Hoang Pham, Yuanlong Wang, **Changchang Yin**, Xueru Zhang, Ping Zhang. Open-set heterogeneous domain adaptation: Theoretical analysis and algorithm. AAAI Conference on Artificial Intelligence (**AAAI**), 2025 (Acceptance rate: 3032/12957 = 23.4%, main track)
6. **Changchang Yin**, Pin-Yu Chen, Bingsheng Yao, Dakuo Wang, Jeffrey Caterino, Ping Zhang. SepsisLab: early sepsis prediction with uncertainty quantification and active sensing. ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**), 2024 (Acceptance rate: 20%)
7. Jiayuan Chen, **Changchang Yin**, Yuanlong Wang, Ping Zhang. Predictive modeling with temporal graphical representation on electronic health records. International Joint Conference on Artificial Intelligence (**IJCAI**), 2024.
8. Shao Zhang, Jianing Yu, Xuhai Xu, **Changchang Yin**, Yuxuan Lu, Bingsheng Yao, Melanie Tory, Lace M Padilla, Jeffrey Caterino, Ping Zhang, Dakuo Wang. Rethinking human-AI collaboration in complex medical decision making: A case study in sepsis diagnosis. ACM CHI Conference on Human Factors in Computing Systems (**CHI**), 2024 (Acceptance rate: 1060/4028 = 26.3%)
9. Yuanlong Wang, **Changchang Yin**, Ping Zhang. Multimodal risk prediction with physiological signals, medical images and clinical notes. *Heliyon* 10:e26772, 2024 (Impact factor: 4.0)
10. Seungyeon Lee*, **Changchang Yin***, Ping Zhang. Stable clinical risk prediction against distribution shift in electronic health records. *Patterns* 4:100828, 2023 (Impact factor: 6.5)
11. Thai-Hoang Pham, **Changchang Yin**, Laxmi Mehta, Xueru Zhang, Ping Zhang. A fair and interpretable network for clinical risk prediction: A regularized multi-view multi-task learning approach. *Knowledge and Information Systems (KAIS)* 65:1487–1521, 2023
12. **Changchang Yin**, Ruoqi Liu, Jeffrey Caterino, Ping Zhang. Deconfounding actor-critic network with policy adaptation for dynamic treatment regimes. ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**), 2022 (Acceptance rate: 254/1695 = 15.0%)
13. **Changchang Yin**, Sayoko Moroi, Ping Zhang. Predicting age-related macular degeneration progression with contrastive attention and time-aware LSTM. ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**), 2022
14. Thai-Hoang Pham, **Changchang Yin**, Laxmi Mehta, Xueru Zhang, Ping Zhang. Cardiac complication risk profiling for cancer survivors via multi-view multi-task learning. IEEE International Conference on Data Mining (**ICDM**), 2021 (Acceptance rate: 98/990 = 9.9%, regular paper, oral presentation. Selected as one of the best papers and invited to KAIS Journal Special Issue)
15. Zicong Zhang, **Changchang Yin**, Ping Zhang. Temporal clustering with external memory network for disease progression modeling. IEEE International Conference on Data Mining (**ICDM**), 2021 (Acceptance rate: 98/990 = 9.9%, regular paper, oral presentation)

16. Fenglin Liu, **Changchang Yin**, Xian Wu, Shen Ge, Ping Zhang, Xu Sun. Contrastive attention for automatic medical report generation. Findings of Annual Meeting of the Association for Computational Linguistics (Findings of ACL), 2021
17. Yuanfang Guan, Hongyang Li, Daiyao Yi, Dongdong Zhang, **Changchang Yin**, Keyu Li, Ping Zhang. A survival model generalized to regression learning algorithms. Nature Computational Science 1:433–440, 2021
18. Zicong Zhang, Kimerly Powell, **Changchang Yin**, Shilei Cao, Dani Gonzalez, Yousef Hannawi, Ping Zhang. Brain atlas guided attention U-net for white matter hyperintensity segmentation. American Medical Informatics Association Informatics Summit (AMIA Summit), 2021 (Buckeye AI, one of the top solutions for WMH Segmentation Challenge)
19. Dongdong Zhang*, **Changchang Yin***, Katherine Hunold, Xiaoqian Jiang, Jeffrey Caterino, Ping Zhang. An interpretable deep-learning model for early prediction of sepsis in the emergency department. Patterns 2:100196, 2021 (Buckeye AI, one of the winning teams for 2019 DII National Data Science Challenge)
20. Sundreen Asad Kamal*, **Changchang Yin***, Buyue Qian, Ping Zhang. An interpretable risk prediction model for healthcare with pattern attention. BMC Medical Informatics and Decision Making 20:307, 2020
21. Ruoqi Liu, **Changchang Yin**, Ping Zhang. Estimating individual treatment effects with time-varying confounders. IEEE International Conference on Data Mining (ICDM), 2020 (Acceptance rate: 91/930 = 9.8%, regular paper, oral presentation)
22. Dongdong Zhang, **Changchang Yin**, Jucheng Zeng, Xiaohui Yuan, Ping Zhang. Combining structured and unstructured data for predictive models: a deep learning approach. BMC Medical Informatics and Decision Making 20:280, 2020
23. Rui Li*, **Changchang Yin***, Samuel Yang, Buyue Qian, Ping Zhang. Marrying medical domain knowledge with deep learning on electronic health records: a deep visual analytics approach. Journal of Medical Internet Research (JMIR) 22(9):e20645, 2020 (Impact factor: 5.034. Featured on AMIA 2021 Year-in-Review)
24. **Changchang Yin**, Ruoqi Liu, Dongdong Zhang, Ping Zhang. Identifying sepsis subphenotypes via time-aware multi-modal auto-encoder. ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2020 (Acceptance rate: 216/1279 = 16.9%)
25. **Changchang Yin**, Rongjian Zhao, Buyue Qian, Xin Lv, Ping Zhang. Domain knowledge guided deep learning with electronic health records. IEEE International Conference on Data Mining (ICDM), 2019 (Acceptance rate: 95/1046 = 9.1%, regular paper, oral presentation)
26. **Changchang Yin**, Buyue Qian, Xianli Zhang, Yang Li, Qinghua Zheng. Automatic generation of medical imaging diagnostic report with hierarchical recurrent neural network. IEEE International Conference on Data Mining (ICDM), 2019 (Acceptance rate: 95/1046 = 9.1%, regular paper, oral presentation)
27. Xianli Zhang, Buyue Qian, Yang Li, **Changchang Yin**, Xudong Wang, Qinghua Zheng. KnowRisk: an interpretable knowledge-guided model for disease risk prediction. 2019 IEEE International Conference on Data Mining (ICDM), 1492-1497
28. **Changchang Yin**, Buyue Qian, Shilei Cao, Xiaoyu Li, Jishang Wei, Qinghua Zheng, Ian Davidson. Deep similarity-based batch mode active learning with exploration-exploitation. IEEE International Conference on Data Mining (ICDM), 2017.
29. Shilei Cao, Buyue Qian, **Changchang Yin**, Xiaoyu Li, Jishang Wei, Qinghua Zheng, Ian Davidson. Knowledge guided short-text classification for healthcare applications. 2017 IEEE International Conference on Data Mining (ICDM), 31-40
30. Zihao Zhu, **Changchang Yin**, Buyue Qian, Yu Cheng, Jishang Wei, Fei Wang. Measuring patient similarities via a deep architecture with medical concept embedding. IEEE International Conference on Data Mining (ICDM), 2016.

HONORS & AWARDS

- **BMI Excellent Research Award** (Graduate Student), OSU 2024
- **DII National Data Science Challenge** (sepsis onset prediction), Honorable Mention Award 2019

- **Tianchi Lung Nodule Detection Challenge**, 5th place / 3000+ teams 2017
- **National Scholarship**, XJTU (top 1%) 2015

PRESENTATIONS

- **"SepsisLab: early sepsis prediction with uncertainty quantification and active sensing"**
Presentation for KDD'2024 Aug 2024
- **"Deconfounding actor-critic network with policy adaptation for dynamic treatment regimes"**
Presentation for KDD'2022 Aug 2022
- **"Predicting age-related macular degeneration progression with contrastive attention and time-aware LSTM"**
Presentation for KDD'2022 Aug 2022
- **"Identifying sepsis subphenotypes via time-aware multi-modal auto-encoder"**
Presentation for KDD'2020 Aug 2020
- **"Deep similarity-based batch mode active learning with exploration-exploitation"**
Presentation for ICDM'2017 Dec 2017
- **"Measuring patient similarities via a deep architecture with medical concept embedding"**
Presentation for ICDM'2016 Dec 2016

MENTORING

- Jiayuan Chen, Master student 2023 - present
 - Co-authored paper on IJCAI 2024
- Zishan Gu, Ph.D. student 2023 - present
 - Two co-authored submissions on EMNLP 2024 and Cell Reports Methods 2024
- Seungyeon Lee, Ph.D. student, Korean Female 2022 - present
 - Co-authored paper on Patterns 2023
- Yuanlong Wang, Ph.D. student 2022 - present
 - Co-authored paper on Heliyon 2024
- Jana Abedeljaber, Undergraduate, American Muslim Female Summer 2023
 - Neurological recovery prediction project in George B. Moody PhysioNet Challenge 2023
- Zicong Zhang, Master student 2019-2021
 - Two co-authored papers on ICDM 2021 and AMIA 2021
- Dongdong Zhang, visiting Master student 2019-2021
 - Two co-authored papers on Patterns 2021 and BMC Medical Informatics and Decision Making 2020.
- Sundreen Asad Kamal, Master student, Pakistani Muslim Female 2019-2020
 - Co-authored paper on BMC Medical Informatics and Decision Making 2020.
- Xianli Zhang, Ph.D. student 2016-2019
 - Co-authored paper on ICDM 2019

TEACHING EXPERIENCE

- Guest Instructor for BMI 5553 EHR Analytics (Graduate Level) Autumn 2024

- Teaching Assistant for CSE 6249 Trustworthy Data Science (Graduate Level) Spring 2023
- Teaching Assistant for CSE 6539 Artificial Intelligence in Medicine (Graduate Level) Autumn 2022

SERVICES

- Conference reviewer: KDD, ICLR, AAAI, IJCAI, ICDM, SDM, AMIA.
- Journal reviewer: npj, JMIR, JBHI, NCAA, BMC Medical Informatics and Decision Making.
- Conference sub reviewer: ICML, NeurIPS.
- Journal sub reviewer: Nature Communications, Nature Machine Intelligence, Patterns, BMC Bioinformatics.